Muhammad Islam Kamran

Backend Engineer – Python & FastAPI

REST & WebSocket APIs · AI Integrations (OpenAI Whisper, GPT) · MySQL · AWS · Team Lead

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Technical Skills

Backend: Python (FastAPI, Django, Flask), RESTful APIs, WebSockets, SQLAlchemy, Alembic, JWT / OAuth2, Role-Based Access, Caching, Rate Limiting, Cron Jobs
AI & Automation: OpenAI Whisper (speech-to-text), GPT-based applications, LLM APIs, RAG with Pinecone, OCR & Document Bots, Selenium, AI-Assisted Development
Databases: MySQL, SQLite, MongoDB
Cloud & DevOps: AWS (EC2, RDS, Elastic Beanstalk, Load Balancing), PM2, InMotion Hosting, Vercel, Render, GitHub Actions, Git
Tools & Frontend: New Relic, Postman, Google Maps & Places APIs, Angular, TypeScript, JavaScript, React, PHP
Currently Learning: Docker, Apache Kafka, PostgreSQL, PyTest

Professional Experience

Indus Real Estate LLC – Apparel Group **March 2025 – Present**
Team Lead – IT & Innovation *Dubai, UAE (On-site)*

- Lead a team of **4 engineers** delivering the company's web platforms end to end, from planning and API design to deployment and support.
- Built **4 portals** (properties, agents, technical services "Indus Specialty" and the General Manager's portal) with **FastAPI**, **Angular / TypeScript** and **MySQL**.
- Built real-time notifications and live listing sync from **Property Finder** using **WebSockets**.
- Secured APIs with **JWT** authentication and middleware; added caching, rate limiting and scheduled cron jobs.
- Shipped **15+ landing pages**; deploy and run services on **AWS** (EC2, RDS, load balancing, PM2) and InMotion Hosting.

Business Innovation & Tech Solutions (BITS) **May 2024 – Aug 2026**
Lead Backend Engineer & Technical Consultant (part-time from Mar 2025) *Hybrid – Dubai / Pakistan*

- Built **FastAPI** backends for **Big Star** and **Zeeshan Motors** (Baig Group, Dubai) on MySQL and SQLite with SQLAlchemy.
- Integrated **Google Maps**, Places and container APIs; built **Selenium** scraping and automation pipelines.
- Added rate limiting and caching to public APIs and monitored performance with **New Relic**.
- Led backend, frontend and DevOps; deployed on **AWS** EC2 / RDS with load balancing and PM2.

K2X Technologies **May 2023 – April 2024**
Software Engineer / Product Owner *Pakistan (On-site) – clients via SquareX, Denmark*

- **Neptune (GPT-based AI product):** built the **Flask** backend and Google sign-in; maintained on AWS with the cloud team; delivered to client Drip.
- **Rollout:** built the **FastAPI** backend with SQLAlchemy, Alembic and MySQL; **JWT / OAuth2** with role-based access; New Relic monitoring.
- Product owner for the **ECOWAS** energy project (UN SDGs), managing a team of 5; backend and on-site engineering for **DGIPR** (Govt. of KP).

CareCloud (MTBC) **October 2022 – April 2023**
Software Engineer *Pakistan*

- **AI speech:** built **Django** APIs for the speaking component of Rosetta, an English-learning app, using **OpenAI Whisper**.
- Built **Flask** document bots for FOX Caregiver (US healthcare) that extract and score data from medical documents into **MongoDB**.
- Worked on a Flask **OCR** app extracting data from 47 commonly used US insurance cards.

AI Projects & Research

LLM & RAG Applications | *Python, LLM APIs, Pinecone* **Personal**

- Built LLM and retrieval-augmented generation (RAG) apps with vector search (Pinecone) through an applied LLM course; shared on LinkedIn.

Quantum Walk Image Encryption | *MATLAB, Python, Cryptography* **2021 – 2022**

- First-author, peer-reviewed paper in **CMC (Impact Factor 3.7)** from my MPhil research fellowship: image encryption with quantum walks and chaotic maps. Paper

Education

Quaid-i-Azam University (QAU) **2020 – 2022**
MPhil in Computer Science – CGPA 3.70 / 4.00 *Islamabad, Pakistan*

University of Engineering & Technology (UET) **2016 – 2020**
Bachelor of Science (BS) in Computer Systems Engineering *Peshawar, Pakistan*